

Anonymization Through Data Synthesis Using Generative Adversarial Networks (ADS-GAN)

Jinsung Yoon , *Member, IEEE*, Lydia N. Drumright, and Mihaela van der Schaar, *Fellow, IEEE*

Abstract—The medical and machine learning communities are relying on the promise of artificial intelligence (AI) to transform medicine through enabling more accurate decisions and personalized treatment. However, progress is slow. Legal and ethical issues around unconsented patient data and privacy is one of the limiting factors in data sharing, resulting in a significant barrier in accessing routinely collected electronic health records (EHR) by the machine learning community. We propose a novel framework for generating synthetic data that closely approximates the joint distribution of variables in an original EHR dataset, providing a readily accessible, legally and ethically appropriate solution to support more open data sharing, enabling the development of AI solutions. In order to address issues around lack of clarity in defining sufficient anonymization, we created a quantifiable, mathematical definition for “identifiability”. We used a conditional generative adversarial networks (GAN) framework to generate synthetic data while minimize patient identifiability that is defined based on the probability of re-identification given the combination of all data on any individual patient. We compared models fitted to our synthetically generated data to those fitted to the real data across four independent datasets to evaluate similarity in model performance, while assessing the extent to which original observations can be identified from the synthetic data. Our model, ADS-GAN, consistently outperformed state-of-the-art methods, and demonstrated reliability in the joint distributions. We propose that this method could be used to develop datasets that can be made publicly available while considerably lowering the risk of breaching patient confidentiality.

Index Terms—Synthetic data generation, anonymization, identifiability, generative adversarial networks, electronic health records.

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I. INTRODUCTION

THE adoption of electronic health records (EHR) has dramatically increased in high-income countries over the last decade [1]–[4] with corresponding interest to do so in low and middle income countries worldwide [5]. Evidence from both small scale studies and other disciplines suggests that Artificial Intelligence (AI) could support significant advances in health-care delivery [6], [7], however, appropriate legal and ethical management of routinely collected EHR can create obstacles to open sharing of sensitive health data. This is further complicated by domain specific concepts from distinct disciplines or perspectives, such as elevating differential privacy [8] above the concept of risk benefit ratio with respect to societal gains versus individual risk.

In order to mitigate the legal and ethical risks of sharing sensitive information, de-identification of patient records is commonly used, for example creating “shareable” medical datasets such as the MIMIC-III dataset [9]. However, previous work has shown that de-identification is not sufficient for avoiding re-identification through linkage with other identifiable datasets [10]–[13]. Additionally, data protection regulations, such as the General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPAA) do not provide clear guidance on anonymization of personally identifiable data [14], [15], exacerbating concern and complicating appropriate data sharing.

Differential Privacy (DP) [8] has been presented by computer scientists as a potential solution to this problem; however, it has some limitations that may not be fit for purpose when considering healthcare data. Briefly, DP for a positive ϵ , ϵ -differential privacy is defined as follows: For all datasets D_1 and D_2 that differ on a single element, and for all subsets S of the function image of the randomized algorithm $\mathcal{A}$,

$$P(\mathcal{A}(D_1) \in S) \leq e^\epsilon P(\mathcal{A}(D_2) \in S).$$

In other words, if the randomized algorithm $\mathcal{A}$ (here it can be a synthetic data generator) is said to provide ϵ -differential privacy, the probability distribution differences between all possible D_1 and D_2 for all subsets of the function image of the randomized algorithm should be less than e^ϵ .

Data protection in health and medicine is governed by laws, such as those defined by the General Data Protection Regulation (GDPR) and ethics. DP is not a requirement in medical/ health setting, nor is it familiar to those working in this area. There are some technical limitations in DP as well. First, DP cannot

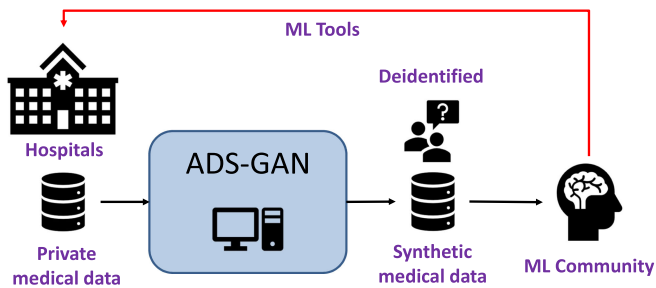


Fig. 1. ADS-GAN enables to share the private (identifiable) data (by sharing de-identified synthetic data) to machine learning community to develop machine learning tools easier.

be directly computed using finite real datasets. It is defined on all possible subsets of the algorithm image with a probability measure that needs an infinite number of samples and an extremely large number of computations. Therefore, DP is just theoretically guaranteed by differentially private procedure of generating synthetic data. In the healthcare domain, the most interesting privacy aspect is to measure the identifiability of the finite original patient data by the finite generated synthetic data, which is impossible in DP. Second, the quality of synthetic data generated by the differentially private method is usually poor with respect to having a similar joint distribution to the original dataset, a critical component of this work, because the theoretical upper bound of DP is loose. Therefore, the performance of the constructed machine learning model with synthetic data generated by the differentially private method is not practically useful in many downstream tasks.

Generating synthetic patient records based on real patient records can be an alternative way of providing machine learning researchers with data that they need to be able to develop appropriate methods for the task at hand, while avoiding sharing sensitive patient information. However, generating realistic synthetic data is challenging in this setting because the patient records are often high-dimensional and come from complex distributions. Moreover, there will often be a relatively small number of patients with unique characteristics (for example, patients with a rare disease or those who are outliers) and it is therefore even harder to accurately estimate the complex, high-dimensional distribution of these data without replicating the individual.

Generative Adversarial Networks (GAN) [16] provide a promising framework for simulating complex distributions. The GAN framework has demonstrated dramatic improvements in synthetic image generation [17], image translation [18], [19], and other applications [20]–[22]. The defining feature of the GAN framework is the existence of a generator and discriminator, trained in an adversarial fashion against each other. The generator tries to generate synthetic samples that the discriminator is incapable of distinguishing from the real samples, while the discriminator is trying to identify which of the samples are the synthetic ones. This framework can be formulated as a minimax game and at the optimal point of this game, generated samples follow the real data distribution.

Our model, ADS-GAN (anonymization through data synthesis using generative adversarial networks), modifies the conditional GAN framework [23] where the generator and discriminator are given additional inputs in the form of the values of a set of conditioning variables. Unlike the standard conditional GAN framework where the set of variables that are conditioned on is pre-determined, ADS-GAN optimizes a conditioning set for each patient and generates all components based on these. We do this to improve the quality of the synthetic data while also ensuring that no combination of features could readily reveal a patient's identity. This also allows us to move away from the differential privacy solution, and concerns about highly improbable re-identification, and define identifiability based on the probability of re-identification given the combination of all data on any individual patient. Finally, there are two well-known limitations in the original GAN framework: (1) The converged generator only generates similar samples (called mode collapse), (2) The training procedures of the generator and the discriminator are unstable because the two components are adversarially trained. To mitigate these two problems, we apply Wasserstein GAN with Gradient Penalty (WGAN-GP) [24], [25] as the base GAN architecture for ADS-GAN instead of the original GAN framework. WGAN-GP utilizes the Wasserstein distance instead of the Jensen-Shannon Divergence (that the original GAN framework is based on) as the measure of distances between the synthetic and real distributions. This can solve the limitation of the Jensen-Shannon Divergence that needs to have the same supports for the probability measure. Furthermore, the training procedure of the WGAN-GP is not adversarial; thus, the training procedure is more stable than the original GAN framework.

A. Motivation - Quantitatively Defining Identifiability, Regulation, Confidentiality and Risk-Benefit Ratio

In computer science the issue of confidentiality has largely been addressed by focusing on differential privacy [8]. In many ways differential privacy solutions make sense given both the lack of definitive definitions for anonymization in policy and law and the potential for risk, however it fails to address key concepts in medical and health research. This creates an unintended barrier to developing AI to advance healthcare and public health delivery, due to an inability to use routinely collected clinical data for AI research and development. We propose to address this in part through both creating a quantifiable mathematical definition of identifiability and through development of a novel method of generating robust synthetic datasets from EHR data.

Current data protection legislation with respect to health records in both Europe (GDPR) and the US (HIPAA), aims to protect the confidentiality of persons and preserve the tacit agreement between them and their health provider, stating that 'personal data' cannot be processed for other purposes, for example health records for research where consent has not been granted. However, these regulations suffer from lack of clarity in the definitions of 'personal data,' and 'anonymization' that render them no longer 'personal data'. The GDPR, which has been adopted by the European Union (EU) and is currently being adopted in

the US, provides the following definition of ‘personal data’ in article 4, “any information relating to an identified or identifiable natural person (‘data subject’); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person;”[14]. HIPAA has similarly unclear definitions [15], [26].

To help address this lack of clarity, different types of data identifiability have been proposed [27]. Direct identifiers, also referred to as obviously identifiable data, include full names, or specific uniquely identifiable numbers assigned to individuals (e.g., social security number, hospital number, driving license number, etc). Indirect identifiers, sometimes termed “likely identifiable data,” have been described as less obvious unique information, such as date of birth, initials, phone numbers, and postcodes. We also proposed that there are two other categories. Potentially identifiable information, information that in the presence of additional data may allow for re-identification of an individual, for example dates of medical procedures or visits, rare conditions, etc. Non-identifiable information including laboratory and physiology test results, clinical coding, etc. There is little controversy around the need to remove obviously and likely identifiable information from a dataset to anonymize it. Potentially identifiable information and combinations of non-identifiable information are our main areas of concern relating to re-identification of a individual in a dataset.

Those expressing the most extreme concerns around anonymized data postulate that “any information that distinguishes one individual from another can be used for re-identifying anonymous data.”[28]; often ignoring the educational, technological and data access requirements to conduct such sophisticated re-identification, resulting in an extremely low probability of re-identification. Such ideas have led to the ever-mounting focus on differential privacy as a solution, however, differential privacy techniques are not well suited for highly complex medical data in which noise is already inherent and preservation of joint distributions is critical. Differential privacy in the setting of secondary use of routinely collected medical data for understanding of health conditions and developing decision support, also fails to address ethical concerns pertaining to the risk benefit ratio, where minimal risk may be deemed allowable if the societal benefits are high. In the case of using AI to enable delivery of more effective and efficient healthcare, there are significant societal benefits.

Given the complicated dynamics around protecting, anonymizing and sharing routinely collected health data, we decided to address the problem in a new way. Instead of defining personal data or risk, which have both been thus far unclear, we mathematically define identifiability. Moreover, we developed a model to create entirely synthetic datasets of individuals that are fictitious and yet could be drawn from the same population as the real dataset. Within this model the user can define their threshold of acceptable identifiability. This allows the model operator to modulate the risk benefit ratio for any given problem.

II. RELATED WORK

Others have previously attempted to generate synthetic patient “information sets” using summary statistics of original datasets [29] or domain-knowledge “association sets” of information [30]. However, these methods are limited to low-dimensional feature spaces and specific diseases and do not provide the end user with individual level observations required for many models. Choi *et al.* [31] generated synthetic patient records using a GAN framework without any identifiability constraints, which we have formulated for ADS-GAN. Choi and colleagues [31] focused only on generating discrete variables, whereas ADS-GAN is capable of generating mixed-type variables (continuous, discrete, and binary variables). The ADS-GAN model significantly improves the quality of the generated synthetic samples by conditioning on the optimized selected components of the real patient records. The quantitative experiments in Choi’s study [31] focused only on binary variables, whereas we compare with Choi [31] in both the continuous and the discrete settings and our model significantly outperforms theirs [31] in all cases. DP-GAN [32] and PATE-GAN [33] generate differentially private data using GANs. However, both methods must introduce additional noise over the original GAN framework to satisfy the differential privacy constraint. Therefore the quality of the generated samples, in terms of similarity to the original dataset is likely to be worse than the generated samples by the original GAN framework [16], [24], [25]. On the other hand, ADS-GAN generates the synthetic data conditioned on the original data (i.e. using conditional GAN framework) where conditioning variables are optimized. Therefore, ADS-GAN can generate better quality synthetic data than the original GAN framework even while satisfying the identifiability constraints.

III. PROBLEM FORMULATION

Let us denote the feature space by $\mathcal{X}$. Suppose that X is a random variable over $\mathcal{X}$ distributed according to $\mathcal{P}_X$. We write $\mathbf{x}$ to denote realizations of X and refer to $\mathbf{x}$ as a patient observation with d dimensions. The dataset $\mathcal{D}$ consists of N patient observations, assumed i.i.d. according to $\mathcal{P}_X$: $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i = (x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(d)})$ and $x_i^{(j)} \in \mathcal{X}^{(j)}$ for all $j \in \{1, 2, \dots, d\}$. $x^{(j)}$ represents the j -th feature of the patient observation $\mathbf{x}$ and $\mathcal{X} = \prod_{j=1}^d \mathcal{X}^{(j)}$.

Our goal is to generate a synthetic dataset, $\hat{\mathcal{D}}$, given by $\hat{\mathcal{D}} = \{\hat{\mathbf{x}}_i\}_{i=1}^N$ where each $\hat{\mathbf{x}}$ is drawn from the approximate distribution $\mathcal{P}_{\hat{X}}$, where we want $\mathcal{P}_{\hat{X}}$ to be suitably close to $\mathcal{P}_X$ while ensuring that it is not so close that the identifiability constraint is violated; and the acceptable threshold for identifiability can be modified.

A. Identifiability

Generated synthetic patient observations should be “different enough” from the original patient observations in order to de-identify the original patient observations. We then ask, what is “different enough” from the original observations? We can guarantee that two different observations in the original dataset

are “different enough” because they are “different” patients. Therefore, we use the “minimum” (closest) distance between the original observations as the measure for “different enough” between the synthetic and original data.

The “minimum distance” must then be mathematically defined. However, just using the Euclidean distance does not make sense in terms of identifiability, as some features or values are more rare, and hence more identifiable than others. For example, the distance between male and female patients (50% of patients are male) should be closer than the distance between cancer and non-cancer patients (e.g. 1% of patients have cancer). The value of having cancer is a more rare, and thus more identifiable, value than not having cancer. Therefore, we use the weighted Euclidean distance as the metric to measure the distance between two patient observations in terms of identifiability.

Using the above motivations, we define ϵ -identifiability as follows: there are less than ϵ ratio observations from the original dataset ($\mathcal{D}$) in the generated synthetic dataset ($\hat{\mathcal{D}}$) that are “not different enough” from the original observations. Mathematically, let us define a measurement r_i as follows.

$$r_i = \min_{\mathbf{x}_j \in \mathcal{D}/\mathbf{x}_i} \|\mathbf{w} \cdot (\mathbf{x}_i - \mathbf{x}_j)\|$$

where $\mathcal{D}/\mathbf{x}_i$ represents the dataset $\mathcal{D}$ without $\mathbf{x}_i$ and $\mathbf{w}$ is a weight vector for the weighted Euclidean distance computation. r_i represents the minimum distance between $\mathbf{x}_i$ and the other original observations in $\mathcal{D}$. This is the mathematical definition of “different enough” for the observation $\mathbf{x}_i$. Note that the threshold for “different enough” can be different for different observations. Because “different enough” in an outlier should be much larger (in terms of distance) than “different enough” in the common observation to de-identify the patient information.

Similar to r_i , we define $\hat{r}_i$ as follows.

$$\hat{r}_i = \min_{\hat{\mathbf{x}}_j \in \hat{\mathcal{D}}} \|\mathbf{w} \cdot (\mathbf{x}_i - \hat{\mathbf{x}}_j)\|$$

The definition of $\hat{r}_i$ is the minimum distance between $\mathbf{x}_i$ and the generated observations in $\hat{\mathcal{D}}$. Using r_i and $\hat{r}_i$ we can define ϵ -identifiability as follows.

Definition 1: $\hat{\mathcal{D}}$ is ϵ -identifiable from $\mathcal{D}$ if

$$\mathcal{I}(\mathcal{D}, \hat{\mathcal{D}}) = \frac{1}{N} [\mathbb{I}(\hat{r}_i < r_i)] < \epsilon \quad (1)$$

where $\mathbb{I}$ represents the identity function and

$$r_i = \min_{\mathbf{x}_j \in \mathcal{D}/\mathbf{x}_i} \|\mathbf{w} \cdot (\mathbf{x}_i - \mathbf{x}_j)\|$$

$$\hat{r}_i = \min_{\hat{\mathbf{x}}_j \in \hat{\mathcal{D}}} \|\mathbf{w} \cdot (\mathbf{x}_i - \hat{\mathbf{x}}_j)\|$$

ϵ -identifiability means that for each observation in $\mathcal{D}$, defined as the probability that the distance to the closest synthetic observation is closer (“not different enough”) than the distance from the closest real observation, is less than ϵ . In other words, $\geq 1 - \epsilon$ is the proportion of patients that are not identifiable by synthetic observations because the synthetic patient observations are “different enough” from the real patient observations. Therefore, 0-identifiability would represent a perfectly non-identifiable

dataset and 1-identifiability would represent a perfectly identifiable dataset. The proposed identifiability is defined for all the samples or variables (not for particular samples or a group of variables belonging to an individual patient) and for each sample, the identifiability is differentially determined. For the outlier, we apply harsher constraints (to add more noise) to make it less identifiable. With any synthetic and original datasets, we can compute the identifiability based on the definition 1.

1) Weight Vector: The weight vector $\mathbf{w}$ for the definition of weighted Euclidean distance ($\|\mathbf{w} \cdot (\mathbf{x} - \hat{\mathbf{x}})\|$) can be defined in various ways. The user can directly define the weight vector based on the domain knowledge for the identifiability of each feature. For instance, for non-identifiable features, such as a biochemistry result, users can set $w_i = 0$ and for identifiable features, such as a diagnosis, users can set $w_i = 1$.

In this paper, we use the discrete entropy as the definition of the weight vector,¹ but any other weight definitions can be applied in the proposed framework. The discrete entropy of the feature is defined as follows.

$$H(X^{(i)}) = - \sum_{x^{(i)} \in \mathcal{X}^{(i)}} \mathcal{P}(X^{(i)} = x^{(i)}) \log(\mathcal{P}(X^{(i)} = x^{(i)}))$$

where $H(X^{(i)})$ represents the entropy of the i -th feature. The entropy represents the uncertainty of the feature; therefore, the weights for identifiability should be the inverse of the entropy, because less uncertainty results in greater identifiability. Therefore, we introduce the weight vector ($\mathbf{w}$) for the weighted Euclidean distance metric as follows.

$$\mathbf{w} = (w_1, \dots, w_d) = \left(\frac{1}{H(X^{(1)})}, \dots, \frac{1}{H(X^{(d)})} \right)$$

where $w_i \in [0, \infty)$.

B. Objectives

Our two main objectives in generating a synthetic dataset are: (1) ensuring that the samples from the synthetic dataset are similarly distributed to those in the original dataset, and (2) ensuring the generated samples are sufficiently different (“different enough”) from the original samples which can be measured by ϵ -identifiability (where $\epsilon \in [0, 1]$). It is therefore important to note that, naturally, we have a trade-off between these two objectives - conveying sufficient information about the structure of the original dataset so that the development of machine learning methods on the synthetic dataset approximates the original dataset, while preventing re-identification by limiting information about the real patient. We formalize this using the adversarial learning framework (minimax optimization problem) [16] as follows.

$$\min_{\mathcal{P}_{\hat{\mathbf{x}}}} \max_{\mathcal{D}} [\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X} [\log D(\mathbf{x})] + \mathbb{E}_{\hat{\mathbf{x}} \sim \mathcal{P}_{\hat{X}}} [\log(1 - D(\hat{\mathbf{x}}))]]$$

$$\text{subject to } \mathcal{I}(\mathcal{D}, \hat{\mathcal{D}}) < \epsilon \quad (2)$$

where $\epsilon > 0$ is the identifiability threshold to be specified by the user. This equation can be described as follows: the discriminator

¹We treat all of the features as discrete features with c categories (for continuous features, we quantize, for binary, $c = 2$).

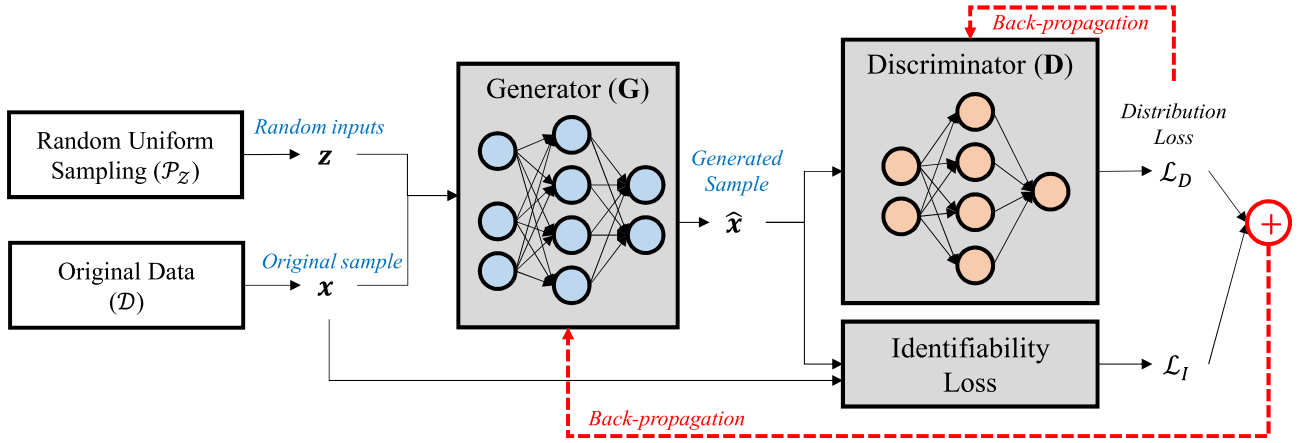


Fig. 2. Block diagram of ADS-GAN. The generator uses (1) original sample ($\mathbf{x}$) and (2) random vector ($\mathbf{z}$) to generate sample ($\hat{\mathbf{x}}$). The summation of distribution loss ($\mathcal{L}_D$) and identifiability loss ($\mathcal{L}_I$) is back-propagated to the generator. Both the generator and discriminator are implemented with multi-layer perceptrons.

(D) tries to distinguish between samples from the original data ($\mathcal{P}_X$) and samples from the synthetic data ($\mathcal{P}_{\hat{X}}$). On the other hand, the generator ($\mathcal{P}_X$) tries to generate synthetic data such that the discriminator cannot distinguish the generated samples from the original samples. More specifically, the objective function can be interpreted as the accuracy of the discriminator to classify the original and generated samples correctly. The discriminator tries to maximize this accuracy while the generator tries to minimize this accuracy, which is represented as the min-max optimization problem. Simultaneously, the identifiability constraint $\mathcal{I}(\mathcal{D}, \hat{\mathcal{D}}) < \epsilon$ is satisfied.

IV. ADS-GAN: ANONYMIZATION THROUGH DATA SYNTHESIS USING GAN

We used the modified conditional Generative Adversarial Networks (coGAN) framework [23] to generate synthetic data that satisfy the identifiability constraints described above. The proposed model (ADS-GAN) consists of two components: the Generator and the Discriminator. The block diagram of ADS-GAN is illustrated in Fig. 2.

A. Generator

The generator tries to generate the synthetic data that has a joint distribution similar to the joint distribution of the original data. Let us define the generator function G as follows $G : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{X}$. We define the synthetic data $\hat{\mathbf{x}}$ by

$$\hat{\mathbf{x}} = G(\mathbf{x}, \mathbf{z})$$

where $\mathbf{x} \sim \mathcal{P}_X$ and $\mathbf{z} \sim \mathcal{P}_Z$. $\mathcal{P}_Z$ can be any known distribution such as a multi-variate Gaussian or Uniform distribution. In other words, the generated observation $\hat{\mathbf{x}}$ is sampled by the generator function G conditioning on the original observation $\mathbf{x}$. In this paper, we implement the generator function G using multi-layer perceptron and the parameters for the generator are denoted as θ_G .

In detail, we first use the min-max scaler to normalize the continuous variables to $[0, 1]$ and apply one-hot encoding to the categorical variables so that all the variables are in $[0, 1]$ range. Then, we concatenate normalized continuous variables and one-hot encoded categorical variables and put them in the input layer of the perceptron. At the output layer of the perceptron, we divide the continuous variables and categorical variables and use sigmoid activation to the continuous variables and softmax activation to the categorical variables to conserve the property of the continuous and categorical variables. Lastly, we re-normalize the continuous variables and reverse the one-hot encoded data to the categorical data to make the synthetic data the same format as the original data.

B. Discriminator

The discriminator in ADS-GAN is defined as a function $D : \mathcal{X} \rightarrow \mathbb{R}$. The discriminator D is trained to measure Wasserstein distances between the joint distributions of the generated observations ($\hat{\mathbf{x}}$) and the real observations ($\mathbf{x}$). Based on the Wasserstein GAN-GP framework (WGAN-GP) [24], [25], the loss is given by $\mathcal{L}_D$

$$[\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X, \hat{\mathbf{x}} \sim \mathcal{P}_{\hat{X}}} [D(\mathbf{x}) - D(\hat{\mathbf{x}}) - \eta(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2]]$$

where $\gamma \sim \text{Uniform}(0, 1)$ and $\tilde{\mathbf{x}} = \gamma\mathbf{x} + (1 - \gamma)\hat{\mathbf{x}}$ and η is the hyper-parameter (in this paper we assigned a value of 10 to hyper parameter γ based on previous work [25]). The maximum value of the loss ($\mathcal{L}_D$) is the Wasserstein distance between the original and the synthetic data distributions. Therefore, the role of the discriminator is to measure the Wasserstein distance between the original data distribution and the synthetic data distribution. A large $\mathcal{L}_D$ means that the difference in the joint distributions between the original and synthetic datasets is large. Therefore, we also call this as the distribution loss. In this paper, we implement the discriminator function D using multi-layer perceptron and the parameters for the discriminator are denoted as θ_D .

C. Identifiability Loss

$\mathcal{L}_D$ is only utilized to optimize the objective function of the optimization problem because it is only utilized to minimize the Wasserstein distance between the joint distributions of original and synthetic datasets. It does not guarantee that the identifiability constraint in the optimization problem is satisfied. To ensure that the generated samples satisfy the identifiability constraint as well, we need to introduce the identifiability loss $\mathcal{L}_I$. The straightforward identifiability loss can be defined as

$$\tilde{\mathcal{L}}_I = \max \left(0, \frac{1}{N} [\mathbb{I}(\hat{r}_i < r_i)] - \epsilon \right) \quad (3)$$

However, in order to compute $\tilde{\mathcal{L}}_I$, we need to compute $r_i = \min_{\mathbf{x}_j \in \mathcal{D}/\mathbf{x}_i} \|\mathbf{w} \cdot (\mathbf{x}_i - \mathbf{x}_j)\|$ and $\hat{r}_i = \min_{\hat{\mathbf{x}}_j \in \hat{\mathcal{D}}} \|\mathbf{w} \cdot (\mathbf{x}_i - \hat{\mathbf{x}}_j)\|$ for every observation, which has extremely high computational complexity.² Therefore, we introduce the identifiability loss function $\mathcal{L}_I$ as follows.

$$\mathcal{L}_I = \mathbb{E}_{\mathbf{x}, \hat{\mathbf{x}}|\mathbf{x}} [-\|\mathbf{w} \cdot (\mathbf{x} - \hat{\mathbf{x}})\|] \quad (4)$$

It represents the generator's efforts to maximize the weighted Euclidean distance between generated sample $\hat{\mathbf{x}}$ and the conditioned original sample $\mathbf{x}$ given that $\hat{\mathbf{x}}$ is "different enough" from the original sample $\mathbf{x}$. $\|\mathbf{w} \cdot (\mathbf{x} - \hat{\mathbf{x}})\|$ can be interpreted as the approximation of $\hat{r}_i$ because $\hat{\mathbf{x}}$ is generated from $\mathbf{x}$; thus, there is a high probability that $\hat{\mathbf{x}}$ is the closest data point to $\mathbf{x}$.

D. Optimization

The components of ADS-GAN are adversarially optimized against each other. The generator tries to minimize the loss ($\max_D [\mathcal{L}_D]$) which represents the Wasserstein distance between the original data and the synthetic data distributions. Furthermore, the generator also tries to minimize the identifiability loss ($\mathcal{L}_I$) such that the generated samples are "different enough" from the original samples. Therefore, the final optimization problem can be written as follows.

$$\begin{aligned} \min_G \left[\max_D [\mathcal{L}_D] + \lambda \mathcal{L}_I \right] \\ = \min_G \left[\max_D \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X, \mathbf{z} \sim \mathcal{P}_Z} [D(\mathbf{x}) - D(G(\mathbf{x}, \mathbf{z})) \right. \\ \left. - \eta(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}})\|_2 - 1)^2] - \lambda \mathbb{E}_{\mathbf{x}, \hat{\mathbf{x}}} [\|\mathbf{w} \cdot (\mathbf{x} - \hat{\mathbf{x}})\|] \right] \end{aligned} \quad (5)$$

where $\lambda \geq 0$ is a hyper-parameter. When $\lambda = 0$, the optimization problem is reduced to the original WGAN-GP optimization problem given by

$$\begin{aligned} \min_G \max_D \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X, \mathbf{z} \sim \mathcal{P}_Z} [D(\mathbf{x}) - D(G(\mathbf{x}, \mathbf{z})) \\ - \eta(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}})\|_2 - 1)^2] \end{aligned} \quad (6)$$

In this case, one of the optimal generators is $G^*(\mathbf{x}, \mathbf{z}) = \mathbf{x}$ such that it just outputs the input feature. In this case, the above

²For each computation for r_i we need to run the k-Nearest Neighbor algorithm [34] for N samples that have high computational complexity.

Algorithm 1: Pseudo-Code of ADS-GAN.

Initialize: θ_G, θ_D

Inputs: η, λ

while training loss has not converged **do**

 Draw n mini-batch samples from $\mathcal{D}$, $\{\mathbf{x}_k\}_{k=1}^n$

 Draw n mini-batch samples from $\mathcal{P}_Z$, $\{\mathbf{z}_k\}_{k=1}^n$

for $k = 1, \dots, n$ **do**

$\hat{\mathbf{x}}_k \leftarrow G(\mathbf{x}_k, \mathbf{z}_k)$

end for

(1) Update D with fixed G

 Update θ_D using stochastic gradient ascent

$$\nabla_{\theta_D} \frac{1}{n} \sum_{k=1}^n [D(\mathbf{x}_k) - D(\hat{\mathbf{x}}_k) - \eta(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}}_k)\|_2 - 1)^2]$$

 where $\gamma \sim \text{Uniform}(0, 1)$ and $\tilde{\mathbf{x}}_k = \gamma \mathbf{x}_k + (1 - \gamma) \hat{\mathbf{x}}_k$

(2) Update G with fixed D

 Update θ_G using stochastic gradient descent

$$\begin{aligned} \nabla_{\theta_G} \frac{1}{n} \sum_{k=1}^n [D(\mathbf{x}_k) - D(\hat{\mathbf{x}}_k) - \eta(\|\nabla_{\tilde{\mathbf{x}}} D(\tilde{\mathbf{x}}_k)\|_2 - 1)^2 \\ - \lambda \|\mathbf{w} \cdot (\mathbf{x}_k - \hat{\mathbf{x}}_k)\|] \end{aligned}$$

 where $\gamma \sim \text{Uniform}(0, 1)$ and $\tilde{\mathbf{x}}_k = \gamma \mathbf{x}_k + (1 - \gamma) \hat{\mathbf{x}}_k$

end while

problem is changed to

$$\max_D [\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X} [\log(1 - D(\mathbf{x}))]] \quad (7)$$

therefore, the generator can minimize its loss as $\log(4)$. But the identifiability is 1; meaning that all of the original samples are identifiable.

On the other hand, if $\lambda = \infty$, the proposed optimization problem is converted to the following:

$$\min_G \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X, \mathbf{z} \sim \mathcal{P}_Z} [\|\mathbf{w} \cdot (\mathbf{x} - G(\mathbf{x}, \mathbf{z}))\|] \quad (8)$$

In this case, the optimal generator is $G^*(\mathbf{x}, \mathbf{z}) = \hat{G}(\mathbf{z})$ where $\hat{G}$ is any function from $\mathcal{Z}$ to $\mathcal{X}$. Then, there is no relationship between $\mathbf{x}$ and $\hat{\mathbf{x}}$ and the generator no longer needs to generate synthetic data that follows a similar distribution to that of the original data distribution $\mathcal{P}_X$.

As can be seen from the above two extreme cases, λ adjusts the trade-off between conveying sufficient information about the structure of the original dataset and avoiding providing too much information about the structure such that it would make patients identifiable. If λ is high/low, the proposed method penalizes more/less on the identifiability loss ($\mathcal{L}_I$). Intuitively, with large/small λ , the corresponding identifiability ($\mathcal{I}(\mathcal{D}, \hat{\mathcal{D}})$, represented in Equation (1)) should be small/large. We solve the proposed optimization problem (minimax problem) iteratively, and the proposed generator learns how much noise it should add to the original data in order to satisfy both identifiability and the same distribution constraints. The pseudo-code of ADS-GAN is illustrated in Algorithm 1.

TABLE I
META DATA CHARACTERISTICS OF THE FOUR REAL-WORLD DATASETS

Datasets	MAGGIC	UNOS		
		Heart Transplant	Lung Transplant	Heart Wait-list
Number of Patients	30,389	56,822	26,854	23,706
Number of Features	29	48	41	23
-Continuous	9	15	17	4
-Binary	20	33	24	19
Label ($y = 1$)	5,723 (18.8%)	8,406 (14.8%)	5,523(20.6%)	(48.1%)
Average Correlation between features	0.0758	0.0534	0.0462	0.0589
Average Correlation with labels	0.0584	0.0253	0.0228	0.0627
Max Correlation between features	0.7168	0.8081	0.8745	0.9494
Max Correlation with labels	0.1945	0.1155	0.0962	0.1495

V. EXPERIMENTAL SETTINGS

We divide the original dataset into two disjoint subsets: a training set and a testing set. More specifically, we randomly select 80% of the patients to be used in the training set and the remaining patients are used to form the test set. The synthetic dataset is also randomly divided into a training and a testing set, using the same 80% training and 20% testing division. We evaluate the generated synthetic datasets in terms of various metrics on the testing set.

We compare ADS-GAN with the following benchmarks: PATE-GAN [33], DP-GAN [32], MedGAN [31], and WGAN-GP [25]. The first two benchmarks (PATE-GAN and DP-GAN) generate differentially private synthetic datasets. Therefore, we adjust the differential privacy constraints to control the identifiability levels of the generated synthetic datasets. The last two benchmarks (MedGAN and WGAN-GP) have no identifiability constraints; thus, we cannot adjust the identifiability levels of the generated synthetic datasets. For ADS-GAN, we control the identifiability levels using the hyper-parameter λ that adjusts the trade-off between the quality of the generated data and the identifiability. Note that the identifiability of all generated synthetic data can be directly computed using the definition 1.

A. Data Description

1) *MAGGIC Dataset*: The Meta-analysis Global Group in Chronic Heart Failure (MAGGIC) [35] is a collection of 30 different datasets from 30 different medical studies containing patients who experienced heart failure. We set the label of each patient as 1-year all-cause mortality, excluding all patients who were censored before 1-year (i.e. whose follow-up is less than an year). There were a total of 29 features (binary: 20, continuous: 9) and 30,389 patients in this dataset. Among 30,389 patients, 5,723 (18.8%) patients died within 1 year.

2) *UNOS Dataset*: The United Network for Organ Transplantation (UNOS) dataset (available at <https://www.unos.org/data/>) provides information about all patients in the U.S. who have received an organ transplant or were on the waiting-list during the period 1985–2015. In all 3 datasets (Heart-Transplant, Lung-Transplant, and Heart-Wait-list), the objective is to predict

1-year all-cause mortality. Table I illustrates meta data characteristics.

VI. EXPERIMENTAL RESULTS

A. Comparison on Feature Distributions

In Table II, we compare the original dataset and generated synthetic datasets for MAGGIC using statistical tests for each feature. We use the Student T-test for continuous variables, and Pearson Chi-square test for binary variables. For ADS-GAN, we adjust the identifiability constraints and for PATE-GAN and DP-GAN, we adjust the differential privacy constraint to experimentally set the identifiability level to 0.1. (Note that 0.1 does not mean that 10% of the observations are identifiable. It means that more than 90% of the observations are not identifiable.) We also experiment on various identifiability levels and report the findings of these in Fig. 3. The identifiabilities of MedGAN and WGAN-GP cannot be modified and are 0.15 and 0.16, respectively.

As can be seen in Table II, ADS-GAN generates better synthetic datasets in comparison to other methods in terms of marginal distribution mismatches determined by statistical tests. In the case of ADS-GAN, among 29 features in the MAGGIC data, only 3 features (6.9%) are statistically different from the original data. The other methods have at least 9 features (31.0%) that are statistically different from the original data (with 0.05 significance level). Similar results were observed using all three UNOS datasets.

B. Preserving the Joint Distribution

In the above subsection, we only evaluate the marginal distribution mismatch for each feature. In this subsection, we show how well the generated synthetic data preserves the “joint” distribution of the original data. To measure the distance between joint distributions of the original dataset and the synthetic datasets, we use the Jensen-Shannon Divergence (JSD) and the Wasserstein distance as the performance metrics.³

³We exclude Kullback-Leibler divergence (KL) because the distance in KL changes based on which distribution is ordered first (e.g. $KL(A,B)$ is different from $KL(B,A)$), this does not satisfy our definition of the “distance”.

TABLE II

COMPARISON OF THE FEATURE DISTRIBUTIONS BETWEEN THE ORIGINAL DATASET AND SYNTHETIC DATASETS (GENERATED BY VARIOUS METHODS) WITH 0.1-IDENTIFIABILITY CONSTRAINT ON MAGGIC DATASET. FOR CONTINUOUS VARIABLES, WE REPORT THE MEAN $\pm$ STD, AND FOR BINARY VARIABLES, WE REPORT THE PROPORTION WITH THE CHARACTERISTIC. * DENOTES THAT THE GENERATED SAMPLES ARE STATISTICALLY DIFFERENT FROM THE ORIGINAL SAMPLES WITH SIGNIFICANCE LEVEL 0.05

Feature Name	Original	ADS-GAN	PATE-GAN	DP-GAN	MedGAN	WGAN-GP
Age	68 $\pm$ 12	68 $\pm$ 18	68 $\pm$ 17	68 $\pm$ 20*	68 $\pm$ 18	68 $\pm$ 15
BMI	27 $\pm$ 5	26 $\pm$ 5	27 $\pm$ 8	27 $\pm$ 9	27 $\pm$ 6*	26 $\pm$ 5
Ejection Fraction	38 $\pm$ 20	39 $\pm$ 16	38 $\pm$ 20	40 $\pm$ 22*	38 $\pm$ 20	39 $\pm$ 18
SBP	132 $\pm$ 24	133 $\pm$ 41	133 $\pm$ 30	133 $\pm$ 43	133 $\pm$ 29	132 $\pm$ 34*
DBP	77 $\pm$ 13	77 $\pm$ 12	78 $\pm$ 22*	77 $\pm$ 28*	77 $\pm$ 18	77 $\pm$ 17
Heart Rate	125 $\pm$ 24	125 $\pm$ 34	125 $\pm$ 36	127 $\pm$ 43*	123 $\pm$ 30*	125 $\pm$ 32
Heart Failure Duration	31 $\pm$ 44	31 $\pm$ 62	31 $\pm$ 71	22 $\pm$ 83*	34 $\pm$ 64*	30 $\pm$ 40*
Creatinine	119 $\pm$ 74	117 $\pm$ 131*	121 $\pm$ 101*	89 $\pm$ 223*	125 $\pm$ 117*	121 $\pm$ 76*
Sodium	139 $\pm$ 4	139 $\pm$ 5*	139 $\pm$ 10	138 $\pm$ 15*	139 $\pm$ 4	139 $\pm$ 9
SOBAR	38.6%	38.8%	38.1%	38.4%	39.3%*	39.2%
SOBOE	70.3%	70.8%	68.9%*	70.0%	69.9%	70.6%
Beta Blocker	31.1%	30.9%	31.1%	30.5%	31.8%*	31.6%
ACEI or ARB	72.9%	72.8%	74.1%*	72.4%	72.5%	73.6%*
Gender (Male)	64.3%	64.1%	63.9%	64.2%	64.4%	63.9%
Diabetes	22.7%	22.8%	22.6%	20.6%*	22.3%*	22.8%
Angina	33.5%	33.6%	32.9%	34.1%	34.0%	33.4%
MI	40.2%	40.5%	40.4%	40.9%*	40.8%	40.6%
Atrial Fibrillation	19.5%	19.9%	17.6%*	19.6%	19.4%	19.4%
Stroke	9.9%	10.0%	9.1%*	9.2%*	10.0%	10.0%
COPD	17.7%	18.2%*	18.3%*	18.3%*	18.7%*	18.1%
Hypertension	40.8%	40.6%	40.8%	40.7%	40.5%	40.2%*
Rales	39.7%	39.9%	39.4%	39.8%	40.1%	40.1%
Ischemic	51.9%	51.8%	50.5%*	51.9%	51.4%	50.6%*
CABG	13.6%	13.8%	13.2%	13.6%	13.4%	12.2%*
PCI	9.3%	9.5%	9.5%	9.3%	10.4%*	10.0%*
LBBS	22.3%	22.4%	21.4%*	24.0%*	22.5%	22.8%
OED	28.2%	28.7%	28.5%	28.2%	28.0%	28.3%
Current Smoker	33.4%	33.1%	33.7%	35.6%*	33.5%	32.9%
NYHA ($\geq$ 2)	89.0%	89.1%	88.5%*	90.0%*	88.0%*	88.5%*

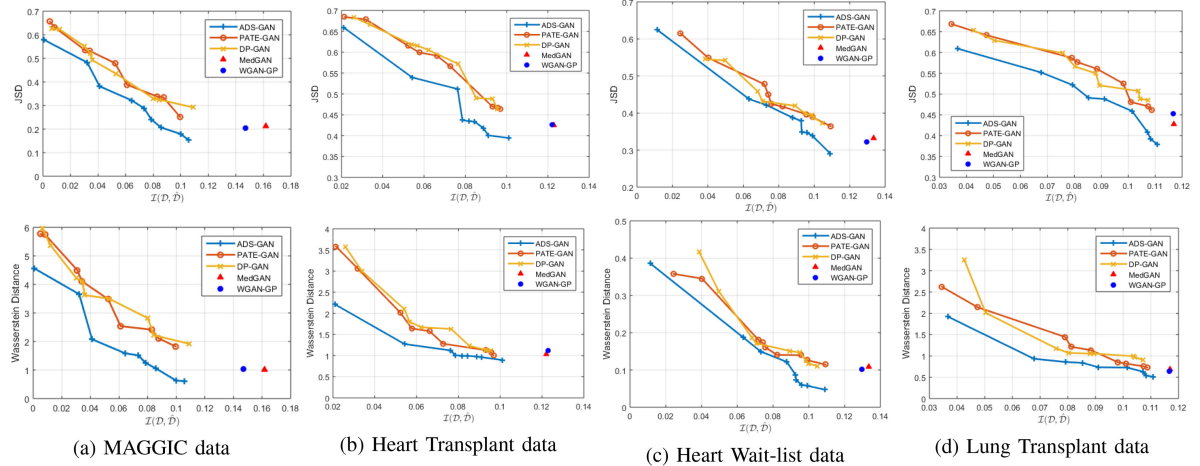


Fig. 3. Trade-off between identifiability and quality of the generated datasets in terms of (first row) JSD, and (second row) Wasserstein distance on Multiple datasets. A lower JSD or Wasserstein distance is better. MedGAN and WGAN-GP have only one point because we cannot control the identifiability levels for those methods.

Jensen-Shannon Divergence ($JSD(\mathcal{P}_X, \mathcal{P}_{\hat{X}})$) and Wasserstein distance ($W(\mathcal{P}_X, \mathcal{P}_{\hat{X}})$) are determined as follows.

$$JSD(\mathcal{P}_X, \mathcal{P}_{\hat{X}}) = \log(2) + \frac{1}{2} \left[\max_g \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X} [\log(g(\mathbf{x}))] + \mathbb{E}_{\hat{\mathbf{x}} \sim \mathcal{P}_{\hat{X}}} [\log(1 - g(\hat{\mathbf{x}}))] \right]$$

$$W(\mathcal{P}_X, \mathcal{P}_{\hat{X}}) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_X} [f(\mathbf{x})] - \mathbb{E}_{\hat{\mathbf{x}} \sim \mathcal{P}_{\hat{X}}} [f(\hat{\mathbf{x}})]$$

where g and f come from any class of functions. We model g and f using neural networks (3-layer perceptron) to approximate the Jensen-Shannon Divergence (JSD) and the Wasserstein distance. If the joint distribution of the synthetic data is similar to the joint distribution of the original data, both JSD and Wasserstein distance have a lower value (if two the joint distributions are the same, both the JSD and Wasserstein distance will be 0). Therefore, in Fig. 3 and Table III, lower values represent better performance.

TABLE III
COMPARISON OF SYNTHETIC DATA GENERATION IN TERMS OF JSD AND WASSERSTEIN DISTANCE WITH 0.1-IDENTIFIABILITY CONSTRAINT ON MULTIPLE DATASETS. (LOWER IS BETTER IN ALL METRICS)

Datasets	MAGGIC		UNOS Heart Transplant		UNOS Heart Wait-list		UNOS Lung Transplant	
Metrics	JSD	Wasserstein Distance	JSD	Wasserstein Distance	JSD	Wasserstein Distance	JSD	Wasserstein Distance
ADS-GAN	0.1835	0.6064	0.3895	0.8991	0.3372	0.0586	0.4621	0.7239
PATE-GAN	0.2514	1.8281	0.4643	1.0014	0.3921	0.1217	0.4825	0.8025
DP-GAN	0.2919	1.9158	0.4612	1.0952	0.3933	0.1175	0.5157	1.0112

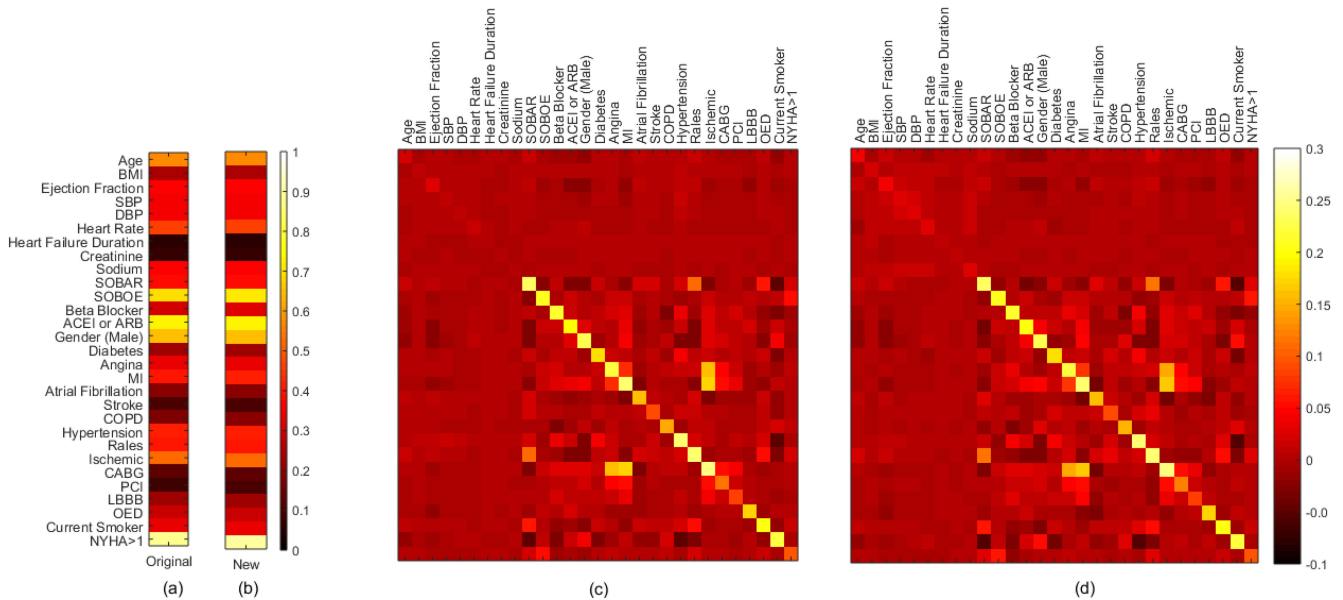


Fig. 4. Visualization on MAGGIC dataset. (a) Mean vectors representations of the original data (b) Mean vectors representations of the generated data by ADS-GAN, (c) co-variance matrix of the original data, (d) co-variance matrix of the generated data by ADS-GAN.

We used the original training set to generate synthetic data. Then, using the original training set and the generated datasets conditioned on the original training set, we trained Multi-layer perceptrons for JSD and Wasserstein distance. We evaluated the JSD and Wasserstein distances on the original testing set and the generated datasets conditioned on the original testing set.

In Table III, we compare ADS-GAN with the benchmarks in terms of JSD and Wasserstein distance. As can be seen in Table 3, ADS-GAN performs significantly better than other benchmarks in terms of all metrics. Lower values for JSD and Wasserstein distance indicate that ADS-GAN preserves the joint distribution of the original data better. For instance, ADS-GAN is 27% and 67% better at preserving the joint distribution than PATE-GAN in terms of both JSD and Wasserstein distance with the same identifiability on the MAGGIC data.

C. Identifiability vs. Performance

In this subsection, we investigate the trade-off between the identifiability and the quality of our synthetic data measured in terms of the JSD and Wasserstein distance. We report the results on both the MAGGIC and the three UNOS Transplant datasets. In the case of MedGAN and WGAN-GP, we cannot adjust the identifiability; thus, we only report one point in Fig. 3.

As can be seen in Fig. 3,⁴ there is a trade-off between the identifiability and the quality of the generated data (represented by JSD and Wasserstein distance). The ADS-GAN model consistently outperforms the benchmarks across the entire range of identifiability levels (note that a lower JSD or Wasserstein distance is better) on both the MAGGIC and the three UNOS Transplant datasets. For instance, with the same quality of the generated samples, ADS-GAN reduced the identifiability by 41% in comparison to WGAN-GP on the MAGGIC dataset; meaning that the ADS-GAN generated samples are 41% less identifiable than the WGAN-GP.

D. Downstream Task Performances With Synthetic Datasets

Among the various applications of synthetic data generation, one of the most important applications is to construct machine learning models using generated synthetic data for various downstream tasks [36]. In this subsection, we evaluate the performance of downstream tasks (3-year mortality predictions) using the generated synthetic data. More specifically, we train the

⁴For DP-GAN and PATE-GAN, the epsilon/delta values are as follows. With the fixed delta as 10^{-5} , we used epsilons in $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10, 50\}$.

TABLE IV
COMPARISON OF SYNTHETIC DATA GENERATION IN TERMS OF
DOWNSTREAM PREDICTION PERFORMANCES (AUROC) WITH
0.1-IDENTIFIABILITY CONSTRAINT ON MULTIPLE DATASETS.
(HIGHER IS BETTER)

Models / Datasets (Metrics: AUROC)	MAGGIC	UNOS Heart Transplant	UNOS Heart Wait-list	UNOS Lung Transplant
Original Data	0.7078	0.8442	0.6376	0.7758
ADS-GAN	0.6929	0.8275	0.6178	0.7536
PATE-GAN	0.6776	0.8165	0.6085	0.7349
DP-GAN	0.6652	0.8137	0.6080	0.7298

machine learning model on the synthetic dataset and evaluate its performance on the original dataset (which we called the *Train on Synthetic, Test on Real* setting). To make the experiments fair, we fixed the identifiability constraint as 0.1 for all the benchmarks and evaluated the performance of downstream tasks (using logistic regression as the predictive model).

As can be seen in Table IV, the proposed method outperforms other benchmarks in all 4 datasets, and the downstream performance in all cases is marginally lower than the performance of the machine learning model that is directly trained on the real data (which is denoted as ‘Original Data’). It also performs better than PATE-GAN or DP-GAN which provides strong evidence that the proposed method outperforms others in terms of generating realistic data.

E. Visualization

In this subsection, we visually compare the mean vectors and co-variance matrices between the original data and the generated data (by ADS-GAN) to show how the generated data follows the joint distribution of the original data on the MAGGIC dataset. We computed mean vector and co-variance matrices for each original and ADS-GAN generated dataset and plot them into the heatmaps. As can be seen in Fig. 4, the heatmaps between original and ADS-GAN data are almost the same, demonstrating that ADS-GAN captures the original joint distribution well.

VII. CONCLUSION

In this paper we proposed a novel framework (called ADS-GAN) for generating synthetic patient records that have similar characteristics to the real patient records while ensuring that a given identifiability constraint is met. We also proposed a measurable definition for the identifiability. We use a modified conditional GAN framework where conditioning variables are not pre-determined but rather are optimized from the real patient records. In all experiments ADS-GAN significantly outperforms all benchmarks in various settings and metrics. We propose that ADS-GAN could be used as a safe, legal and ethical solution to open data sharing of EHR-type datasets, which would support the advancement of AI in Medicine.

APPENDIX A TRUST ASSUMPTION

In this paper, the trust assumption is defined as follows: data guardians will agree on sharing sensitive health data if there

is a certain guarantee on identifiability (i.e. someone cannot re-identify the patient using the shared patient data) and patients will have confidence in this method and it will be viewed as widely acceptable. Based on this assumption, we proposed a synthetic data generation method that satisfies the ethical and legal requirements for protecting patient confidentiality (identifiability) and the distribution constraints (generated synthetic data distribution should be the same with the original data distribution) required to generate quality models that could contribute to the advancement of health and healthcare.

APPENDIX B HYPER PARAMETER OPTIMIZATION AND BENCHMARKS IMPLEMENTATIONS

In all experiments, the depth of the generator and discriminator in both our framework and GAN benchmarks is set to 3. The number of hidden nodes in each layer is d , $d/2$ and d (where d is the feature dimension). We use tanh as the activation function of each layer except for the output layer where we use the sigmoid activation function for the generator and linear activation function for the discriminator. The number of batches is 128 for both the generator and discriminator. The learning rate is $1e-4$ and we use the Adam Optimizer to minimize the loss function. We use tensorflow to implement our model and benchmarks.

APPENDIX C CONNECTION TO k -ANONYMITY

k -Anonymity is a well-known anonymity concept that is defined as follows [37]. The dataset has the k -anonymity property if the information for each person contained in the dataset *cannot be distinguished* from at least $k - 1$ individuals whose information also appears in the dataset. ADS-GAN can be interpreted as generating synthetic dataset that satisfies the 2-anonymity property because there are at least 2 similar data points around one generated synthetic data point due to the identifiability definition that we proposed in this paper. In the case of ADS-GAN, we define *similarity* as the corresponding definition of *indistinguishable* in the k -Anonymity definition.

APPENDIX D THE MINIMUM SIZE OF DATA FOR ADS-GAN

In this paper, we verify the performances of ADS-GAN with 4 different datasets whose number of samples is more than 20,000 (see Table I). Although the quality of generated synthetic data using ADS-GAN depends on both the number of samples and feature dimensions, we discover that with more than 5,000 samples, the overall quality of generated dataset by ADS-GAN is acceptable to use as the synthetic dataset. More specifically, predictive model performances trained with the generated synthetic data by ADS-GAN are similar with the predictive model performances trained with the original data (less than 0.03 differences in terms of AUROC across all 4 datasets).

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